Introduction to Database Systems

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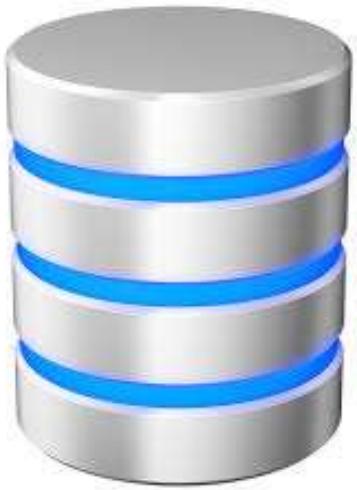
Introduction to databases

TOPICS COVERED IN LECTURE 1

- INTRODUCTION TO DATABASES,
- COMPARISON TO FILE SYSTEM,
- DATABASE ACTORS AND WORKERS,
- ADVANTAGES OF USING THE DBMS APPROACH

Database

- **Database**
  - The collection of data
  - Collection of related data



Properties of Databases

A database has the following implicit properties:

- A database represents some aspect of the real world, sometimes called the **mini world.** Changes to the miniworld are reflected in the database.

- A database is designed, built, and populated with data for a specific purpose. It has an intended group of users and some preconceived applications in which these users are interested.

Some examples

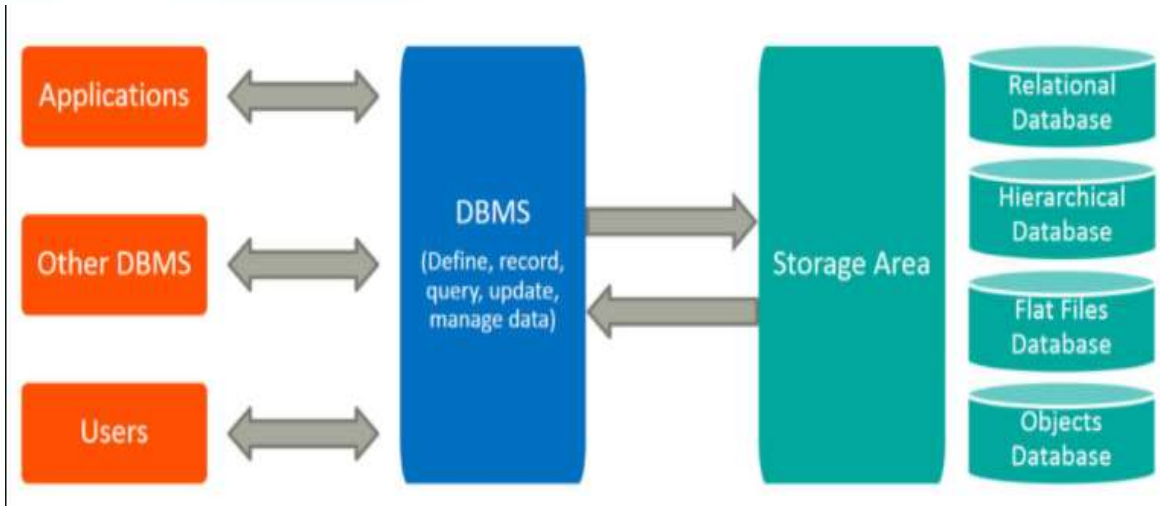
Emergence of web interfaces provided direct database interaction. Data on website is directly fetched from the database.

Examples:

- **Banking:**
  - For customer information, accounts, loans and banking transactions etc.
- **Airlines:**
  - For reservations and schedule information.
- **Universities:**
  - For student information, course registration and grades.
- An example of a large commercial database is Amazon.com

DBMS

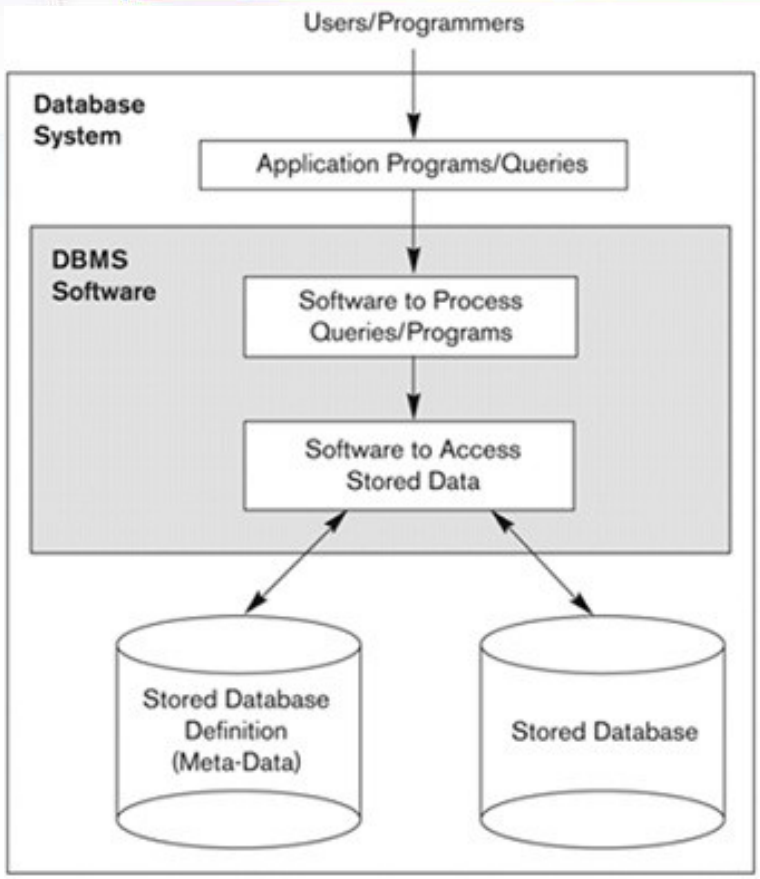
- **Database Management System**
  - Collection of interrelated data and a set of programs to access those data.
- **Goal of DBMS**
  - To provide a way to store and retrieve database information that is both convenient and efficient.



DBMS Functionality (Why to use DBMS)

- **Define** a database in terms of data types, structures and constraints
- **Construct** or Load the Database on a secondary storage medium
- **Manipulating** the database by querying, generating reports, insertions, deletions and modifications to its content
- **Sharing** by a set of users and programs but keeping the data consistent at the same time.
- **Security** measures to prevent unauthorized access.
- **Presentation** and Visualization of data

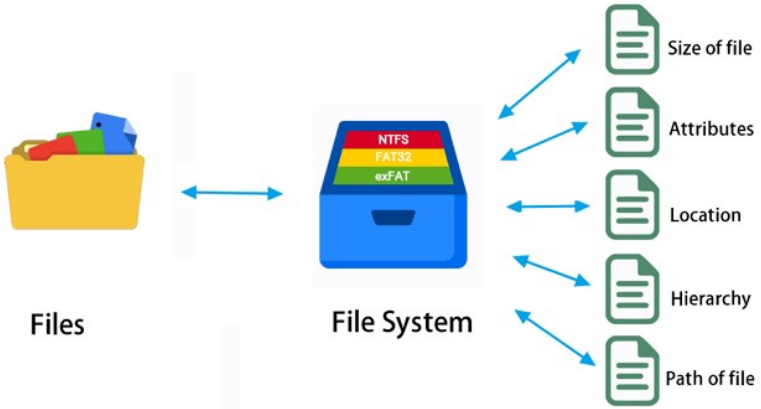
A simplified database system environment



FILE SYSTEM.

File System

- The operating system's file system is a mechanism and data structure for controlling how data is saved and accessed.
- Data deposited in a storage medium without a file system would be one huge body of data with no way of knowing where one piece of data ends and the next begins.
- The data is easily extracted and identified by splitting it into bits and giving each one a name. Each group of data is referred to as a "file".



File System

File System	Database Management System
A file system is a software application that organizes and maintains files on a storage device. It manages the storage and retrieval of data.	A database management system, or DBMS, is a software application that allows you to access, create, and manage databases.
The file system is more related to an operating system and is shipped as a part of the operating system itself.	The database is a software application and is not shipped as a part of the operating system. It is more about organizing the data and implementing techniques to keep the data consistent and to have faster access to the data.
A file system does not allow for efficient data storage and retrieval.	The use of a database management system is efficient because there are numerous techniques for storing and retrieving data.
It does not provide data recovery services.	In DBMS, there is a backup recovery option for data.
There is no crash recovery mechanism in the file system.	A crash recovery method is provided by DBMS.

File System

File System	Database Management System
It's difficult to keep a file system safe.	A good protection method is provided by DBMS.
Support for complex transactions is not available.	Complex transactions are simple to implement.
In the file system, there is no efficient query processing.	In a database management system, data can be retrieved based on the requirements and can be easily filtered from the database.
Concurrency is not available in this system.	A concurrency facility is provided by a database management system.
It is less expensive than a database management system.	It is costlier than that a file system.
It is difficult to implement integrity constraints in file systems.	Integrity constraints can be easily implemented in DBMS.
The system is less complex than a database management system. Complex transactions are not supported by file systems.	As compared to file systems, DBMS is more complex to handle. Complex transactions can be implemented easily in DBMS.

File System

File System	Database Management System
Data independence is not possible in file systems.	DBMS provides data independence.
File systems allow only one user to access data at a time.	Data can be accessed simultaneously by multiple users in a DBMS.
A file system provides less security than a database management system.	DBMSs are more secure than file systems since they have a wider range of security mechanisms.
The data is spread across many files, so sharing it is difficult.	Since data is centralized, it is easy to share.
Data storage and representation details are provided.	By hiding the internal details of the database, it makes it more secure.
Examples – NTFS (New Technology File System), FAT (File Allocation Table), EXT (Extended File System), etc.	Examples – My SQL, Oracle, SQL Server, PostgreSQL, Microsoft SQL Server, Microsoft Access, etc.

ISSUES IN FILLING SYSTEM.

Activity Time

- Lets see an excel sheet holding some records.
- Please see if you find out any problems in the given excel sheet.
- Problems related to :
  - The current state of the file.
  - If you want to add some other rows.
  - If you want to delete some existing rows.
  - If you want to update some information in the existing rows.

University - Microsoft Excel non-commercial use

	Student Id	Student FirstName	Student LastName	Student Address	Student ContactNo	Course Id	Course Name	Course Grades	Instructor Id	Instructor Name	Instructor Address	Instructor ContactNo
2	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV101	Linear Algebra	A	I_01	Paul Koch	9012, rue Royale	34.56.78.90
3	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV204	Visual Basic	B	I_03	Michael Allen	Kirchgasse 9012	(171) 789-0123
4	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV301	Wellness	A	I_02	Mark Hassall	Fauntleroy Circus 4567	(1) 890-1234
5	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV404	Political Science	C	I_10	John Peoples	Cerrito 3456	(5) 456-7890
6	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV501	Network Security	A	I_13	Tom Higgin	Hauptstr. 0123	(11) 012-3456
7	S2	John	Wallker	Avda. de la 5678	(5) 789-0123	UV110	Calculus-2	C	I_41	Venky Krishnan	6789 Baker Blvd.	(503) 555-0122
8	S2	John	Wallker	Avda. de la 5678	(5) 789-0123	UV208	MicroComputer	B	I_23	Shabalin Rostisl	C/ Romero, 1234	(95) 901 23 45
9	S2	John	Wallker	Avda. de la 5678	(5) 789-0123	UV303	Calculus	C	I_12	Jacek Jelitto	Hauptstr. 0123	0452-678901
10	S2	John	Wallker	Avda. de la 5678	(5) 789-0123	UV404	Political Science	A	I_10	John Peoples	Cerrito 3456	(5) 456-7890
11	S2	John	Wallker	Avda. de la 5678	(5) 789-0123	UV501	Network Security	A	I_13	Tom Higgin	Hauptstr. 0123	(11) 012-3456
12	S3	Mary	Mosis	Mataderos 7890	(5) 123-4567	UV101	Linear Algebra	A	I_01	Koch, Paul	9012, rue Royale	34.56.78.92
13	S3	Mary	Mosis	Mataderos 7890	(5) 123-4567	UV204	Visual Basic	B	I_03	Michael Allen	Kirchgasse 9012	(171) 789-0123
14	S3	Mary	Mosis	Mataderos 7890	(5) 123-4567	UV301	Wellness	A	I_02	Mark Hassall	Fauntleroy Circus 4567	(1) 890-1234
15	S3	Mary	Mosis	Mataderos 7890	(5) 123-4567	UV407	Geography	B	I_20	Tianna Jones	Walserweg 4567	(604) 901-2345

1. Redundancy

- Duplication of data
- Same piece of information present at multiple locations

Student_ID	Name	Contact	College	Course	Rank
100	Himanshu	7300934851	GEU	Btech	1
101	Ankit	7900734858	GEU	Btech	1
102	Aysuh	7300936759	GEU	Btech	1
103	Ravi	7300901556	GEU	Btech	1

2. Inconsistency

- Different information related to same entity in different/same files.
- Data can be inconsistent if:
  - Changes are made to one place and not in another
  - There is a typo
  - Data is present in different formats
- Data redundancy can cause data inconsistency

student_id	student name	teacher_id	teacher_name
1	A	1	P
2	B	1	P
3	C	2	Q
4	D	2	R

3. Data Isolation & Data Mapping

- Data is scattered in various files and may be in different formats.
  - Writing new programs for different files may be difficult.
- No link between the files
  - Changes in one cannot be reflected in another

IID	LNAME	FNAME	RANK	SALARY	DEPT
201	Gauss	Fred	Prof	55000	Math
301	Joyce	James	Assoc Prof	23000	English

CID	TITLE	SEMESTER	UNITS	LOCATION	TIME	INSTRUCTOR
MATH23	Calculus	Fall 2000	3	MSRI1	MWF 1-2	201
ENG1A	Literature	Fall 2000	5	DWIN2	TTh 3-4	301

4. Integrity Problems

- Age must be under 60
- Account balance must not be negative
- Balance must be 5000 or greater. If this rule changes, changes would be done at every check.

5. Atomicity

- For example: We are transferring some amount to another account and there is a power outage. What will happen?

ADVANTAGES OF USING DBMS APPROACH

Advantages of DBMS Approach

Following are the advantages of a database management system:-

- **Better Data Transfer**
- **Better Data Security**
- **Data inconsistency is minimized**
- **Better decision-making**
- **Increased end-user productivity**
- **Simple**
- **Faster data access**

USERS OF DATABASE SYSTEM

Database Users

Users may be divided

1. **Actors on the Scene :**

   Those who actually use and control the content

2. **Workers Behind the Scene:**

   Associated with the design, development, and operation of the DBMS *software and system environment.* These persons are typically not interested in the database content itself.

Database Users

Actors on the scene

- **Database administrators:** responsible for authorizing access to the database, for co-ordinating and monitoring its use, acquiring software, and hardware resources, controlling its use and monitoring efficiency of operations.
- **End-users:** they use the data for queries, reports and some of them actually update the database content.

Categories of End-users

1. **Casual** : access database occasionally when needed
2. **Naïve or Parametric** : they make up a large section of the end-user population. They use previously well-defined functions in the form of "canned transactions" against the database.

   Examples are bank-tellers or reservation clerks who do this activity for an entire shift of operations.

Categories of End-users

3. **Sophisticated** : these include business analysts, scientists, engineers, others thoroughly familiar with the system capabilities. Many use tools in the form of software packages that work closely with the stored database.
4. **Stand-alone** : mostly maintain personal databases using ready-to-use packaged applications. An example is a tax program user that creates his or her own internal database.

Categories of End-users

5. **Software Engineers:** Determine the requirements of end users, especially naive and parametric end users, and develop specifications for standard canned transactions that meet these requirements.

6. **Application programmers** implement these specifications as programs; then they test, debug, document, and maintain these canned transactions.

Workers Behind the Scene

- **DBMS system designers and implementers**
  - Design and implement the DBMS modules and interfaces as a software package.
- **Tool developers**
  - **D**esign and implement **tools**—the software packages that facilitate database modeling and design, database system design, and improved performance.
- **Operators and maintenance personnel**
  - **They** are responsible for the actual running and maintenance of the hardware and software environment for the database system.

Thank You all!